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RESEARCH ARTICLE

Biosignals Monitoring for Driver Drowsiness Detection Using Deep Neural Networks

JOSE ALGUINDIGUE[1], AMANDEEP SINGH[1], APURVA NARAYAN[2], AND SIBY SAMUEL[1]
[1]Department of Systems Design Engineering, University of Waterloo, Waterloo, ON N2L 3G1, Canada
[2]Department of Computer Science, Western University, London, ON N6A 3K7, Canada

Corresponding author: Amandeep Singh (a82singh@uwaterloo.ca and amandeep.singh@polymtl.ca)

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ABSTRACT Drowsy driving poses a significant risk to road safety, necessitating the development of reliable drowsiness detection systems. In particular, the advancement of Artificial Intelligence based neuroadaptive systems is imperative to effectively mitigate this risk. Towards reaching this goal, the present research focuses on investigating the efficacy of physiological indicators, including heart rate variability (HRV), percentage of eyelid closure over the pupil over time (PERCLOS), blink rate, blink percentage, and electrodermal activity (EDA) signals, in predicting driver drowsiness. The study was conducted with a cohort of 30 participants in controlled simulated driving scenarios, with half driving in a non-monotonous environment and the other half in a monotonous environment. Three deep learning algorithms were employed: sequential neural network (SNN) for HRV, 1D-convolutional neural network (1D-CNN) for EDA, and convolutional recurrent neural network (CRNN) for eye tracking. The HRV-Based Model and EDA-Based Model exhibited strong performance in drowsiness classification, with the HRV model achieving precision, recall, and F1-score of 98.28%, 98%, and 98%, respectively, and the EDA model achieving 96.32%, 96%, and 96% for the same metrics. The confusion matrix further illustrates the model's performance and highlights high accuracy in both HRV and EDA models, affirming their efficiency in detecting driver drowsiness. However, the Eye-Based Model faced difficulties in identifying drowsiness instances, potentially attributable to dataset imbalances and underrepresentation of specific fatigue states. Despite the challenges, this work significantly contributes to ongoing efforts to improve road safety by laying the foundation for effective real-time neuro-adaptive systems for drowsiness detection and mitigation.

INDEX TERMS Road safety, drowsiness, simulated driving, heart rate variability, electrodermal activity, eye tracking, deep learning algorithms.

I. INTRODUCTION

Drowsiness poses a serious threat to drivers, as it significantly impairs their abilities. This includes increased reaction times for sudden maneuvers, reduced attention levels, and compromised decision-making [1], [2]. Such impairments substantially raise the risk of accidents and fatalities on the road. Despite ample evidence linking fatigue to poor driving performance, effective fatigue detection systems remain limited [3]. This need becomes more critical with the increasing number of drivers and longer commutes. Traditional fatigue detection methods, such as tracking vehicle movements or relying on driver self-reports, have proven inadequate and unreliable [4], [5]. They often lack sensitivity to early signs of fatigue or suffer from subjectivity, as drivers may not accurately assess their fatigue levels due to biases or underestimation of risks. Multiple methodologies exist for assessing drowsiness across diverse domains, categorized into five primary modalities: subjective, physiological, vehicle-based behavioral, hybrid [6] etc. Presently, subjective assessments represent a

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prevalent means of evaluating an individual's level of sleepiness, typically relying on self-reported questionnaires and electrophysiological indices of sleep. Prominent among these tools are the Fatigue Assessment Scale (FAS), a 10-item metric designed to measure symptoms of chronic fatigue [7], the Epworth Sleepiness Scale (ESS), an eight-item self-assessment tool gauging proclivity towards dozing off in non-stressful circumstances [8], and the Stanford Sleepiness Scale (SSS), which employs a seven-point scale for individuals to rate their current alertness level at regular intervals throughout the day [9]. In addition, Karolinska Sleepiness Scale (KSS) is a 9-point scale that measures subjective sleepiness at a particular time, with scores ranging from 1 (extremely alert) to 9 (extremely sleepy, fighting sleep) [10]. Physiological methodologies offer enhanced precision and reliability in drowsiness detection, leveraging the subject's physiological state to discern levels of alertness. These techniques exhibit promising efficacy, manifesting early alterations indicative of drowsiness onset [11]. Commonly employed physiological modalities include Electrocardiogram (ECG), capturing heart muscle electrical activity to derive parameters like heart rate and cardiac intervals [12], and Electroencephalogram (EEG), which monitors brain activity across different frequency bands to discern states of vigilance [11]. Additionally, Electrooculogram (EOG) records potential differences between the cornea and retina, although its invasiveness limits its widespread use [13]. Behavioral assessments entail the observation and analysis of patterns in an individual's behavior during drowsy states, with a particular emphasis on facial expressions such as frequent blinking, nodding, or yawning [6]. These methodologies, often employing video-based systems and machine learning algorithms, aim to detect and interpret such behavioral cues. Notably, while yawning is recognized as a common marker of drowsiness, its reliability is compromised by its occurrence in contexts beyond mere sleepiness [14]. Conversely, eye-state detection techniques like Percentage of Eye Closure (PERCLOS), blink percentage (BP), and blink rate (BR) offer more objective measures, with PERCLOS endorsed by regulatory bodies like the National Highway Traffic Safety Administration (NHTSA) [15]. Hybrid approaches amalgamate multiple measurement modalities to harness their respective strengths and mitigate weaknesses, thereby enhancing the overall efficacy of drowsiness detection systems [6]. Such integrated methodologies often combine physiological, behavioral, and subjective indices to achieve a more comprehensive assessment of sleepiness. In the present study, a hybrid method integrating PERCLOS, BP, BR, Heart Rate Variability (HRV), and Electrodermal Activity (EDA) will be employed, augmented by pre- and post-study evaluations using the FAS to ascertain participants' fatigue levels.

A. EYE-TRACKING FOR DROWSINESS DETECTION

The utilization of eye-tracking features such as PERCLOS, BP, and BR. PERCLOS, has gained significant attention due to its endorsement by regulatory bodies like the NHTSA and its established efficacy in drowsiness assessment [16], [17], [18], [19]. PERCLOS computation entails dividing the number of instances where the eyes are closed beyond specified thresholds (typically 70% or 80% pupil coverage, denoted as P70 or P80 respectively, or utilizing the Eyelid Closure Rating, EM) by the total number of recorded images within a predefined time interval, multiplied by 100. The resultant ratio serves as an indicator of drowsiness, with PERCLOS values exceeding 0.2 often classified as indicative of a drowsy state [16], [17]. Some studies introduce an intermediate "Questionable state" to enhance classification accuracy. While P80 is recommended by the NHTSA for its superior correlation with drowsiness, the choice between P70, P80, or EM remains contentious. Nonetheless, PERCLOS presents certain limitations, including susceptibility to lighting conditions, variability in eye morphology, and computational demands for real-time analysis [20]. Despite these drawbacks, its integration into dashboard cameras renders it a non-intrusive solution for fatigue detection. BP serves as another pivotal metric in drowsiness detection paradigms, predicated on the notion that changes in blinking patterns correspond to fluctuations in alertness levels. Longer blink durations are commonly associated with heightened drowsiness or fatigue [21], with thresholds for drowsiness typically set at a proportion of closed-eye frames exceeding 30% [22]. BR, meanwhile, emerges as a dynamic feature reflective of drowsiness levels, with studies reporting both increases and decreases in BR under drowsy conditions [23]. However, inconsistencies in BR alterations across studies underscore the multifactorial nature of this metric, with factors such as time of day, activities undertaken, and sleep deprivation exerting notable influences [24]. Furthermore, contextual nuances must be considered when interpreting BR data within the framework of drowsiness detection, given its susceptibility to situational variations. Consequently, the establishment of universal BR thresholds for drowsiness detection necessitates careful consideration of these contextual elements and population-specific characteristics.

B. HRV FOR DROWSINESS DETECTION

HRV has emerged as a promising avenue for drowsiness detection, addressing a significant factor contributing to vehicular accidents and associated fatalities [25]. By scrutinizing HRV signals, which reflect the dynamics of the Autonomic Nervous System (ANS), researchers aim to discern patterns indicative of drowsiness, characterized by alterations in ANS activity [16], [26]. Leveraging HRV for drowsiness assessment offers notable advantages, including its non-invasive nature, ease of measurement, and sensitivity to changes in ANS functioning [27]. HRV analysis encompasses both time and frequency domain parameters, with alterations in these domains serving as key indicators of drowsiness. In the time domain, metrics such as the root mean square of successive differences (RMSSD) exhibit

increases attributed to heightened parasympathetic activity during drowsy states [24]. Similarly, in the frequency domain, the high-frequency (HF) band typically increases while the low-frequency (LF) band decreases, reflecting parasympathetic dominance associated with drowsiness [28]. Consequently, HRV analysis offers a robust framework for detecting drowsiness in drivers, with particular emphasis on changes in RMSSD and HF band parameters. While HRV-based drowsiness detection methods have shown promise, certain limitations warrant consideration. Factors such as illuminance and color temperature can potentially influence HRV measures, necessitating careful control in experimental settings [29]. Moreover, some HRV-based drowsiness detection approaches may require supplementary information, such as ECG-derived respiratory data or facial images, to enhance accuracy [30], [31].

C. EDA FOR DROWSINESS DETECTION

EDA has gained attention as a potential marker of drowsiness, with applications spanning diverse research endeavors [32], [33]. EDA measurement encompasses both endosomatic and exosomatic methodologies, focusing on changes in skin conductance indicative of arousal levels [32]. Notably, EDA signals tend to exhibit instability during drowsy states, rendering them potentially valuable indicators of sleepiness [33]. Despite its historical association with lie detection, EDA has demonstrated utility in assessing stress levels and, by extension, conditions predisposing individuals to sleep disorders and cardiovascular ailments [33]. EDA data can be acquired non-invasively during routine activities, circumventing the need for controlled laboratory environments. However, while EDA can distinguish between wakefulness and sleep states, its efficacy in identifying specific sleep stages remains inconclusive [34]. Additionally, during deep sleep phases, EDA signals may present as elevated high-frequency components, posing challenges for accurate interpretation. Despite its potential in drowsiness detection, research on implementing EDA as a feature for detecting driving-related drowsiness remains limited, necessitating further investigation to ascertain its efficacy in this context. Thus, while EDA holds promise as a supplementary measure for drowsiness assessment, its applicability in vehicular safety contexts warrants scrutiny and validation.

D. ARTIFICIAL INTELLIGENCE ALGORITHMS FOR DROWSINESS DETECTION

In recent years, the integration of Artificial Intelligence (AI) algorithms into drowsiness detection systems has garnered considerable attention, offering a proactive approach to mitigating the risks associated with driver fatigue. One notable approach involves fatigue driving detection, which relies on facial multi-feature fusion, amalgamating various facial characteristics to discern signs of fatigue [19]. This method leverages machine learning models like the YOLO-V3 convolutional neural network to capture each facial region. Additionally, the Dlib toolkit is employed to introduce the Eye Feature Vector (EFV) and Mouth Feature Vector (MFV), which assess the states of the eyes and mouth, respectively. Subsequently, metrics such as closed eyes time, blink frequency, and yawn frequency are calculated to evaluate the driver's fatigue state. Researchers employing this approach have achieved a high accuracy of 95.10% in detecting driver fatigue. Another notable advancement is the development of a real-time comprehensive algorithm based on drivers' facial expressions to detect fatigue [17]. This algorithm utilizes a deep convolutional neural network (DCNN) to extract features such as the eye aspect ratio (EAR), mouth aspect ratio (MAR), and PERCLOS from video recordings based on facial landmarks. The overall accuracy of this system has been reported at 95.1%. Similarly, another study utilized the driver's facial features for real-time fatigue detection [16]. The algorithm employed in this study consists of three key components: facial feature location, state recognition, and fatigue state estimation. Initially, multitask convolutional neural networks (MTCNNs) and practical Facial Landmark Detector (PFDL) classification networks are utilized to extract features from the eyes and mouth. Subsequently, a sliding window model is applied to assess the driver's fatigue state by combining metrics such as PERCLOS, blinking frequency, and yawning frequency. This approach achieved an impressive accuracy of 95% in detecting drowsiness, with an average detection time of 38.72 ms. Moving beyond facial features, studies have explored the integration of physiological signals, such as PERCLOS, with advanced analytical techniques. For instance, a study utilized a multivariate multiscale entropy (MMSE) method to differentiate between drowsy, tired, and awake states, achieving a classification accuracy of 76.2% [35]. Moreover, the utilization of PERCLOS in conjunction with the Stanford Sleeping Scale (SSS) facilitated the determination of fatigue thresholds, resulting in a recognition rate of 97.8% across various fatigue levels [36]. Furthermore, the incorporation of deep learning techniques has significantly enhanced drowsiness detection systems. Adaptive neural networks based on MobileNet-V2 and ResNet-50V2 have been employed to detect blink patterns, achieving an impressive accuracy of 97% [37]. Additionally, an algorithm utilizing high frame rate video analysis and fuzzy logic demonstrated over 80% accuracy in drowsiness detection, requiring no adjustments and offering independence from driver-specific settings [38]. In efforts to enhance the robustness of drowsiness detection systems, studies have integrated the LF/HF ratio from HRV analysis alongside PERCLOS, achieving an accuracy rate of 92.5% [18]. Similarly, utilizing HRV features such as SDNN, RMSSD, LF, and HF, in conjunction with backpropagation neural networks (BPNN) and long short-term memory (LSTM) models, yielded true positive rates of 75% and accuracies of 88% [39]. Moreover, some studies have explored the potential of utilizing HRV analysis alone for drowsiness detection, achieving impressive accuracies of up to 99.94% using deep learning methods

such as one-dimensional convolutional neural networks (1D CNNs) [25]. Additionally, neural network-based approaches focusing on HRV features such as HF, LF, and the LF/HF ratio have attained accuracies of 90% [40]. While HRV and PERCLOS have been extensively studied for drowsiness detection, Electrodermal Activity (EDA) has also garnered interest in some studies. A hybrid model capturing EDA, respiration (RESP), and Photoplethysmography (PPG) data achieved an average accuracy of 97% [33]. Additionally, an EDA-based wearable device was developed to detect drowsiness in individuals [41]. EDA was selected as the variable to capture, as it is related to sweat, which directly corresponds to the subject's mental state. Although no machine learning or additional studies were conducted to create a drowsiness detection system, the author demonstrated the relationship between EDA activity in a participant's active, asleep, and drowsy states.

Our primary objective is to create a real-time drowsiness detection system tailored for road driving scenarios, particularly relevant to autonomous and conditionally automated vehicles. This phase of study evaluates the effectiveness of physiological indicators such as HRV, PERCLOS, BR, BP, and EDA signals in predicting driver drowsiness. Leveraging deep learning algorithms like sequential neural networks (SNN), 1D-convolutional neural networks (1D-CNN), and convolutional recurrent neural networks (CRNN), we develop models for analyzing HRV, EDA, and eye tracking data. Evaluation metrics including precision, recall, F1-score, and confusion matrix analysis are employed to assess model performance. This study seeks to advance road safety through the development of efficient real-time neuroadaptive systems for drowsiness detection and intervention. We aim to compare drowsiness detection in monotonous and non-monotonous driving conditions using a multi-modal approach. By analyzing physiological and behavioral data across these distinct scenarios, we seek to enhance the understanding and effectiveness of drowsiness detection systems in varied driving environments.

II. METHODOLOGY

A. PARTICIPANTS

A sample of 33 participants were recruited for the study. However, data from 3 participants was found to be incomplete, resulting in the presentation of findings based on data from 30 participants. Participants eligible for the research were required to be over 18 years of age and hold a valid Canadian Driver's License (e.g., Ontario Class G2 or G) with at least one year of driving experience. Moreover, they had to be active drivers during the time of the experiment. Participants were screened using a cut-off score of 23 on the MSSQ-Short [42], and those lacking full or corrected eye vision were excluded from the experiment. Participants were randomly selected to participate in either the monotonous or non-monotonous scenario. The group chosen for the non-monotonous scenario served as a control group and was compared against the monotonous participants. Recruitment occurred following ethical approval (ORE#44726) to conduct the study, through channels such as social media, flyers, and email lists. The participant pool primarily comprised students from the University of Waterloo, as well as neighboring universities and colleges. The selected participants had driving experience between 2000 and over 30000 kilometers on average per year. The demographic information and other preliminary collected information have been presented in Table 1.

TABLE 1. Participant characteristics.

Characteristics	Monotonous	Non-Monotonous
Age range (years)	20-30	18-43
Age (Mean ± SD years)	24.53 ± 3.46	25.86 ± 6.36
Males	9	11
Females	6	4
G-driving license	7	11
G2- driving license	8	5
Driving experience range (years)	1-13	
Mean driving experience (years)	4.93	8.23
Most consumed product	Coffee	Coffee
Average drowsiness level while drive (1-10)	3.79	4

B. EXPERIMENTAL SETUP AND STUDY DESIGN

The driving scenarios were developed using CARLA [43], an open-source autonomous driving simulator. The simulation and eye tracking software were installed on a computer equipped with an AMD Ryzen 9 5950Xi processor, 32 GB of RGB RAM (4 × 8 GB), an NVIDIA GeForce RTX 3080 LHR Graphics Card, and GIGABYTE AORUS Gen 4 7000s M.2 1TB SSD. The setup also included a Logitech G29 steering wheel controller with force feedback and a set of pedals for vehicle control. The driving scenarios were displayed on a single 27-inch Full HD 1080p LED monitor, providing participants with a high-resolution and wide-angle view of the road ahead. Two distinct driving scenarios: monotonous and non-monotonous were used as illustrated in Figure 1. Participants were randomly assigned to one of these scenarios. To maintain a level of realism, each scenario included AI vehicles on the road, primarily composed of sedans and SUVs. This was achieved through a script designed to spawn a specific set of vehicle types. In the monotonous scenario, 50 vehicles were present on the road, as opposed to 100 in the non-monotonous scenario. Although a certain level of realism was aimed for in the monotonous scenario, the number of vehicles on the road was kept to a minimum to avoid causing significant distraction. The non-monotonous scenario featured trees and surrounding environment close

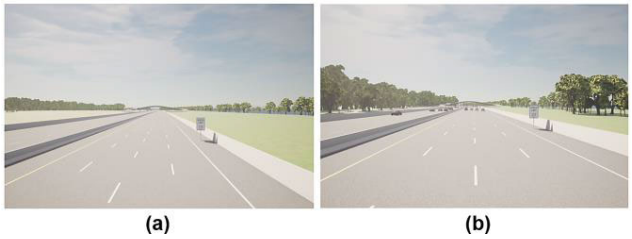

FIGURE 1. Illustration of a) monotonous and b) non-monotonous driving scenario.

to the road, whereas in the monotonous scenario, the environment was dispersed and kept at a distance from the road to minimize potential distractions. Participants drove a blue Dodge Charger for 45 minutes at a speed between 90 km/h and 120 km/h on a four-lane closed loop highway of 8.1 km long. To ensure consistent conditions across both groups, the road conditions, climate, and the highway itself remained the same for both scenarios, with a sunny, clear day prevailing in each. Speed limit signs were posted every 900 meters as stated on [6], indicating a maximum speed of 110 km/h.

The study implemented a mixed factorial design (2 × 2 × 2; scenario type × driving license type × daily caffeine consumption), with age as a covariate. The independent variables between subjects included scenario type (monotonous vs non-monotonous), driving license type (G vs G2), and daily caffeine consumption (yes vs no). Additionally, the study conducted multiple within-subject measurements related to drowsiness. These measurements comprised the PERCLOS, BP, BR, HRV, and EDA. Notably, all these metrics were continuously tracked during the experiment.

C. INSTRUMENTAION

This study used Ergoneers Dikablis Glasses 3 eye-tracker and the Empatica E4 wristband as shown in Figure 2. The Ergoneers Dikablis 3 is a wearable wired binocular eye-tracker with a 60 Hz eye camera tracking frequency, 0.05° visual angle pupil tracking accuracy, and 0.1°- 0.3° visual angle glance direction accuracy. It includes an eye camera with a resolution of up to 648 × 488 pixels and a field camera with a resolution of 1920 × 1080 pixels at 30 fps (frames per second). Calibration is performed to ensure accurate tracking by having the participant fixate on markers placed at four corner points of the screen. The eye-tracker also captures the wearer's pupils using a red circular crosshair. The Empatica E4 wristband combines EDA and PPG sensors, allowing simultaneous measurement of sympathetic nervous system activity and heart rate at a sampling rate of 64Hz. The PPG sensor measures Blood Volume Pressure (BVP), Interbeat Interval (IBI), and heart rate (HR). The GSR sensor, also known as the EDA sensor, detects fluctuating changes in certain electrical properties of the skin. The wristband also includes a 3-axis accelerometer to capture motion-based activity and a skin temperature sensor. The device connects via Bluetooth to a smartphone, where the data is stored in the cloud and automatically synchronized before recording.

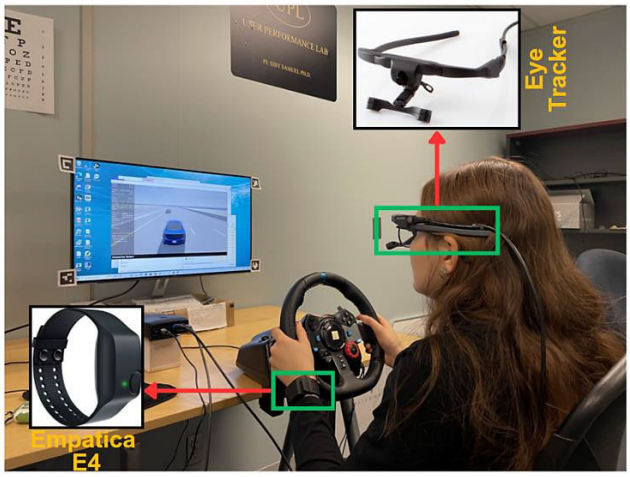

FIGURE 2. Demonstration of instrumentation mounted on the participant during experimentation.

D. DEPENDENT, INDEPENDENT AND CONTROL VARIABLES

The dependent variable for this investigation is the "Level of Drowsiness" in participating subjects. Drowsiness is quantified using a three-tiered ordinal scale: "0" indicating" No Drowsiness," "1" indicating" Possible Drowsiness," and "2" indicating" Definite Drowsiness." This classification is made based on a combination of established physiological and visual indicators, including specific values and patterns in the independent variables. The independent variables are eye tracking features (PERCLOS, BP and BR), HRV features (RMSSD, LF and HF) and EDA. To ensure the validity of the study, two control variables have been identified: 1) Caffeine Consumption: As caffeine is a known stimulant that can impact physiological indicators associated with drowsiness, all subjects will be instructed to abstain from consuming caffeine-related products on the day of the test; 2) Driving Scenario: To control for environmental influences on drowsiness, the driving scenarios used in the simulator will be classified as either 'Monotonous' or 'Non-Monotonous'. This classification will be accounted for during the data analysis phase to control for its potential effects on the state of drowsiness.

E. PROCEDURE

Before participating in the study, all participants provided informed consent. They completed a brief demographics questionnaire, providing personal information such as age and driving experience, along with the Motion Sickness Susceptibility Questionnaire Short-Form (MSSQ-Short) [26]. Additionally, we gathered data on driving experience, sleep patterns, and caffeine consumption habits. It included questions about average yearly kilometers driven, weekly driving duration, caffeine intake, and typical work hours. After completing the questionnaires, participants filled out the FAS and an Activity Log. Following this, participants adjusted their

seat positions as needed, and the Ergoneers Dikablis Glasses 3 eye-tracker and Empatica E4 were fitted and calibrated for each participant. Before starting the experiment, participants were briefed on the simulator and its controls. A preliminary test run on the driving simulator allowed participants to familiarize themselves with the controls and driving conditions while baseline HRV data were recorded. Once participants felt comfortable and confident with the driving task and controls, typically within 5-10 minutes of driving, the main phase of the experiment began. This involved a 45-minute drive under specific scenarios (either monotonous or non-monotonous). Throughout both the test run and the actual experiment, participants were instructed to maintain a speed close to the posted limit of 110 km/h. Participants in the monotonous group were additionally instructed to stay in their lane for the duration of the test. At the end of the experiment, participants completed the FAS for a second time to enable comparative analysis of their results. Each participant received thanks for their contribution, appropriate compensation, and a verbal debriefing.

F. DATA PREPROCESSING

Initially, the BVP and IBI data, gathered from the Empatica E4 device, were transformed into HRV segments lasting 1.5 minutes (90 seconds) each. This conversion was facilitated through a Python script specifically crafted for this purpose. The process commenced with the application of a bandpass Butterworth filter to mitigate noise and continued with successive differences, culminating in a frequency-domain HRV analysis. The subsequent computational steps involved the extraction of time-domain HRV features (RMSSD, SDNN, pNN50) from the filtered BVP readings, followed by the determination of frequency-domain attributes (VLF, LF, HF, LF/HF) using the Welch method for power spectral density estimation. After the HRV data transformation, a baseline normalization was conducted using the initial 5 minutes of participant recordings as they acquainted themselves with the system. The data was then labelled using a threshold method to gauge the drowsiness levels. The thresholds were established based on the following criteria: RMSSD at 1.5, HF at 140, and LF at -1. We labeled as: Label 0: None of the thresholds were met, indicating alertness; Label 1: Any one of the thresholds was met, signaling potential drowsiness; and Label 2: Two or more thresholds were met, suggesting a definitive state of drowsiness or sleepiness. The EDA data was segmented into 90-second intervals, with various parameters such as mean, standard deviation, and range being extracted. The data underwent baseline normalization and labelling based on specific thresholds: mean (1), standard deviation (0.05), and range (1). For the eye data, a pupil detection system integrated into the eye tracker software was utilized, ensuring a minimum accuracy rate of 95% in pupil detection for each participant. This system generated a CSV file for each session, documenting metrics like pupil length, height, and area. The pupil area data facilitated the creation of labels indicating the state of the eyes (open or closed). Moreover, PERCLOS was computed at intervals of 30 seconds, while BP and BR were calculated every 60 seconds. The drowsiness levels were determined using a threshold system with the following criteria: PERCLOS (8), BP (30), and BR (20). It was labeled as: Label 0: No thresholds were met, indicating alertness; Label 1: Any one of the thresholds was met, indicating potential drowsiness; and Label 2: Two or more thresholds were met, indicating a definitive state of drowsiness. To synchronize with the eye tracker software's recording rate of approximately 130 fps, the labels underwent down sampling to match the video sampling rate of 60 fps. This process entailed the removal of even rows from the CSV file, given that the calculated variables remained constant over time frames of at least 30 seconds, yielding a single value for each 30-second label interval. Lastly, the frames were resized to 64×64 pixels prior to their integration into the model.

G. PREDICTION MODELING

A SNN model with multiple hidden layers was selected for HRV data (time-series numeric values). This model can analyze non-linear relationships, recognize patterns, and make accurate predictions on unseen data. It excels in capturing temporal patterns in sequential data. For EDA data (time-series numeric values), a 1D-CNN model was utilized to capture temporal patterns and identify potential drowsy scenarios. It handles sequential data well and can analyze local patterns related to drowsiness. For eye image data, a CRNN was employed to extract spatial features from images and consider temporal dynamics. The dataset is split into training and testing sets using scikit-learn's train test split function, with 80% assigned to the training set (X train and y train) and 20% to the testing set (X test and y test). Following training, all models undergo evaluation using common performance metrics. For the HRV and EDA models, accuracy is assessed on the test dataset by comparing predicted class labels with actual labels, generating a confusion matrix to provide insights into model performance. Additionally, precision, recall, and F1-score are calculated for a detailed evaluation. Evaluation of the Eye model involves metrics such as accuracy, precision, recall, and loss, with a confusion matrix for analyzing predictions on the test data. In this research, hyperparameter tuning was manually conducted. A variety of configurations were tested on the data, with their performance assessed based on pertinent evaluation metrics. For example, within the HRV model, various numbers of layers and neurons were tested, leading to the observation that the optimal architecture comprised five hidden layers with 64, 32, 16, 8, and 4 neurons respectively. Similarly, the hyperbolic tangent (tanh) function was chosen for the input layer owing to its superior performance, while the rectified linear unit (ReLU) was selected for the hidden layers in terms of activation functions. In the EDA model, a sequence of Conv1D layers with differing filter sizes and quantities were examined. The conclusion was drawn that a filter size of 3 with 64 filters produced the most beneficial results.

Additionally, the dropout rate was adjusted, revealing the optimal value to be 0.5 For the Eye model, the number of LSTM layers and the quantity of units in each layer were adjusted. After thorough testing, it was decided to employ three LSTM layers, each with 128 units. The dropout rates varied, with 0.25 being pinpointed as the optimal value for both the encoder and intermediate LSTM layers, and 0.2 for the decoder LSTM layer. It is important to note that the chosen configurations were selected based not only on their immediate performance but also their ability to offer effective generalization and avoid overfitting. This ensures that the models are well-suited for practical deployment in real-world scenarios.

III. RESULTS AND DISCUSSION

A. HRV-BASED MODEL FOR DROWSINESS DETETCION

The HRV model leverages features, specifically RMSSD, HF, and LF, to categorize driver drowsiness. The HRV-Based Model was trained and evaluated using a dataset that consists of 30 CSV files, yielding a combined total of 867 samples, consisting of 669 data points with label 0, 165 with label 1, and 33 with label 2. The dataset underwent preprocessing, where the input features (RMSSD, HF, and LF) and the corresponding output labels (drowsy, alert, etc.) were separated. To facilitate categorical classification, the output labels were subjected to one-hot encoding. For HRV model, Figure 3 presents the training and validation accuracy response over the 40 epochs. Training accuracy commences just below 80%, suggesting that the model initially possessed limited knowledge about the data. A sharp increase in training accuracy is observed in the early stages of training, reaching above 85% by approximately the 5th epoch. This rapid improvement indicates the model's swift learning and adaptation to the training data. Subsequent to the initial sharp rise, the training accuracy continues to improve at a more gradual pace, plateauing around 90%. This behavior demonstrates diminishing returns from learning as the model approaches saturation of its learning capacity from the provided training data. In contrast, the validation accuracy exhibits greater volatility compared to the training accuracy. After the initial epochs, it generally trends higher than the training accuracy, reaching up to approximately 95%. Around the 25th epoch, the validation accuracy attains its peak and then tends to decrease slightly, while the training accuracy continues to improve. This divergence can be indicative of overfitting, where the model becomes excessively tailored to the training data, potentially compromising its generalizability to new, unseen data. By the end of the 40 epochs, the validation accuracy slightly narrows down closer to the training accuracy, suggesting some level of stabilization.

Initially, both the training and validation loss curves exhibit a sharp decline, starting at around 1.0 and rapidly decreasing to approximately 0.6 and 0.65, respectively, by the 5th epoch (refer to Figure 4). This initial sharp drop indicates that the model is quickly learning from the training dataset, effectively capturing the major patterns present in the data. As the training progresses, the loss curves continue to decrease at a slower rate, eventually converging and stabilizing. The training loss stabilizes around 0.3 by the 40th epoch, while the validation loss levels off at around 0.35. This convergence of both losses at a low level suggests that the model has achieved a stable understanding of the data, and further training may provide diminishing returns in terms of loss reduction. Notably, the gap between the training and validation loss curves remains relatively small throughout the training process. Initially, the gap is approximately 0.05, and as training progresses, it widens slightly but remains consistent at around 0.05 by the 40th epoch. This consistent small gap indicates that the model is not overfitting significantly, maintaining good generalization to unseen data, as evidenced by the parallel trajectories of the training and validation loss curves. The observed loss curve behavior demonstrates the model's ability to effectively learn from the training data while maintaining generalization to unseen data. The rapid initial learning, followed by convergence and stabilization of losses, suggests that the model has captured the underlying patterns in the data. The small gap between training and validation losses throughout the training process indicates minimal overfitting, which is a desirable characteristic for robust model performance.

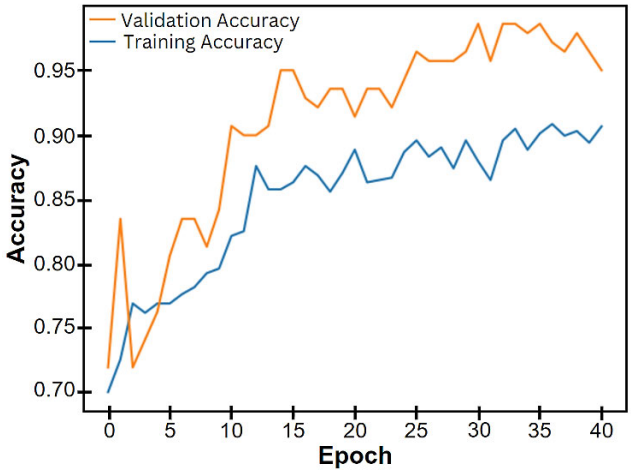

FIGURE 3. Accuracy of HRV-Model over epochs.

Figure 5 presents the performance metrics (i.e., precision, recall and F1-score) for HRV model. The precision score of 98.28% indicates a high accuracy in the positive predictions made by the model. Specifically, it suggests that 98.28% of the HRV instances classified by the model as exhibiting certain characteristics i.e., drowsiness was indeed correctly classified. The recall score of 98% reflects the model's capability to identify all relevant instances of the target HRV patterns. This high recall value implies that the model successfully detected 98% of all actual cases of the specific HRV patterns it was trained to recognize.

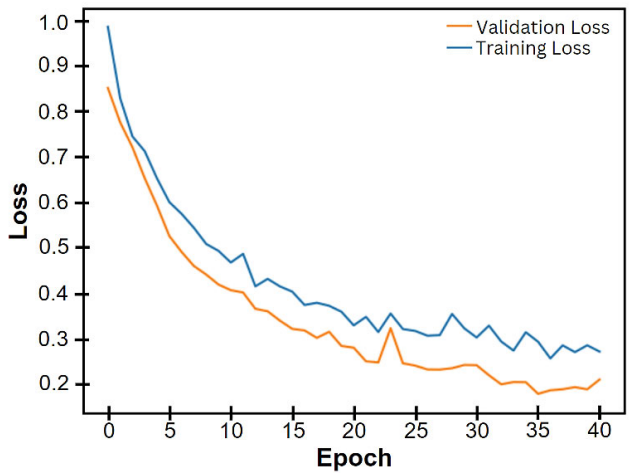

FIGURE 4. Training and validation loss over epochs.

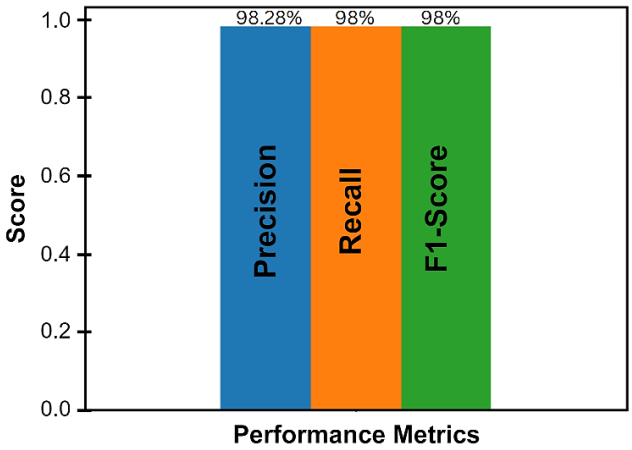

FIGURE 5. Performance metrics for the HRV-Model.

The F1-score, a harmonic mean of precision and recall, stands at 98%, suggesting a well-balanced model with high performance across both accuracy and completeness. The high scores across all three-performance metrics indicate that the HRV model exhibits exceptional performance, demonstrating its effectiveness in accurately recognizing the correct HRV patterns. The confusion matrix visualizes the performance of a HRV classification model across three classes 0, 1, and 2 as shown in Figure 9. For class 0, the model exhibits excellent classification performance, with 131 instances correctly predicted as belonging to this class. Notably, there is minimal confusion with other classes, as only one instance of class 0 was incorrectly predicted as class 1, and one instance from class 1 was misclassified as class 0. This high true positive rate and low false positive/negative rates indicate that the model is highly effective in identifying instances belonging to class 0. Regarding class 1, the model demonstrates good classification accuracy, with 34 instances correctly predicted as class 1. However, there is some confusion with both neighboring classes (0 and 2), as one instance was missed as class 1 and predicted as class 2, one instance as class 0, and two instances from classes 0 and 2 were incorrectly classified as class 1. This suggests that the model may have difficulty distinguishing certain instances of class 1 from the other classes, potentially due to overlapping features or similarities in the HRV patterns. For class 2, despite the low number of instances, the model shows a perfect true positive identification rate, with all six instances correctly predicted as class 2. Remarkably, there were no false negatives for this class, indicating that the model effectively captured all instances belonging to class 2. However, one instance from class 1 was incorrectly classified as class 2, contributing to a false positive.

B. EDA-BASED MODEL FOR DROWSINESS DETETCION

Figure 7 illustrates the training and validation accuracy response of EDA model over 50 epochs. The training accuracy commences at approximately 82.5%, indicating the model's initial performance on the training data. It rapidly ascends to around 90% by the 10th epoch, demonstrating swift learning in the early stages of training. Subsequently, the training accuracy gradually increases and stabilizes around 92.5% by the 50th epoch, suggesting diminishing returns as the model approaches saturation of its learning capacity on the training data. Regarding the validation accuracy, it begins slightly lower than the training accuracy at around 80%, which is expected as the model has not yet learned to generalize to unseen data. However, the validation accuracy increases sharply to about 93% by the 10th epoch, indicating the model's ability to generalize to the validation set. It then fluctuates between 93% and 95%, eventually stabilizing close to 95% by the 50th epoch, showcasing consistent performance on unseen data. Notably, both training and validation accuracies converge, with the validation accuracy occasionally surpassing the training accuracy. This unusual pattern may suggest that the validation set could be slightly easier for the model to predict, or it could be influenced by specific data splits or regularization techniques that help the model generalize better on unseen data. The close tracking of training and validation accuracy with minimal divergence suggests effective generalization, as the validation accuracy does not decrease over time relative to the training accuracy, indicating no significant overfitting. Furthermore, the stabilization of both accuracies towards the later epochs indicates that additional training beyond 50 epochs might not yield significant improvements without changes in model complexity or training data.

Figure 8 illustrates the training and validation loss of EDA model over 50 epochs. Both training and validation loss exhibit a steep decrease initially, indicating that the model is quickly learning from the training data. The training loss decreases from approximately 1.0 to about 0.6 within the first 5 epochs. The validation loss mirrors the training loss with a steep drop from roughly 1.0 to 0.65 in the same interval. This rapid improvement is typical in the early stages of training as the model adjusts its parameters significantly from an

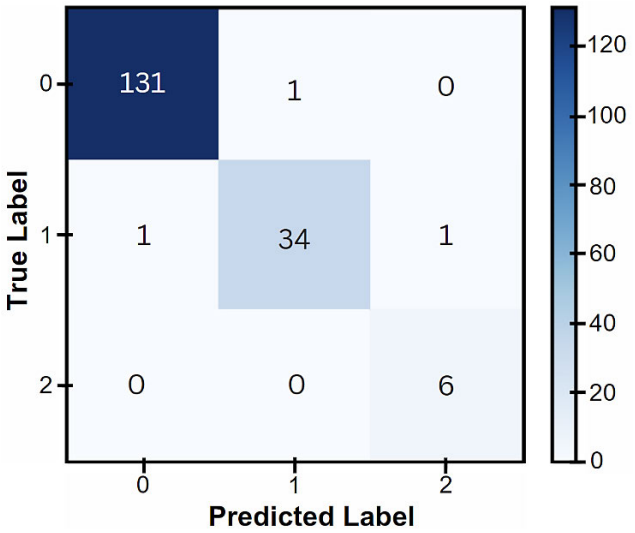

FIGURE 6. Confusion matrix for the HRV-Model.

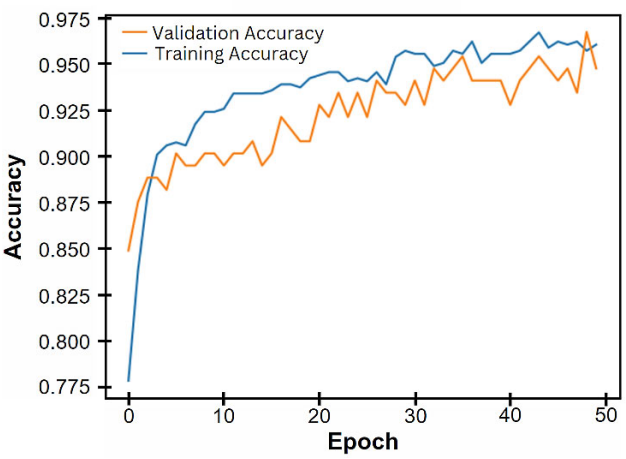

FIGURE 7. Accuracy of EDA-Model over epochs.

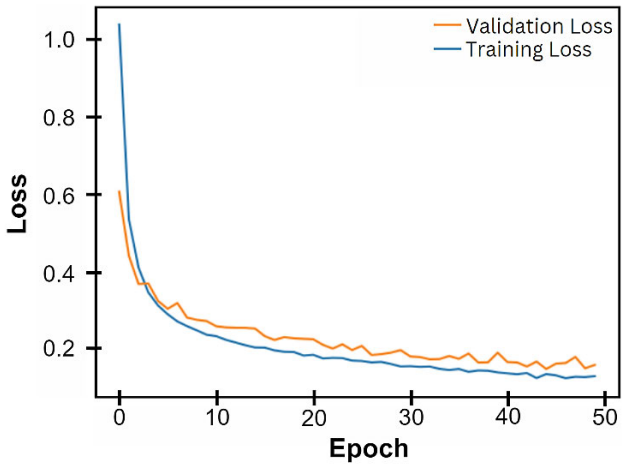

FIGURE 8. Training and validation loss over epochs.

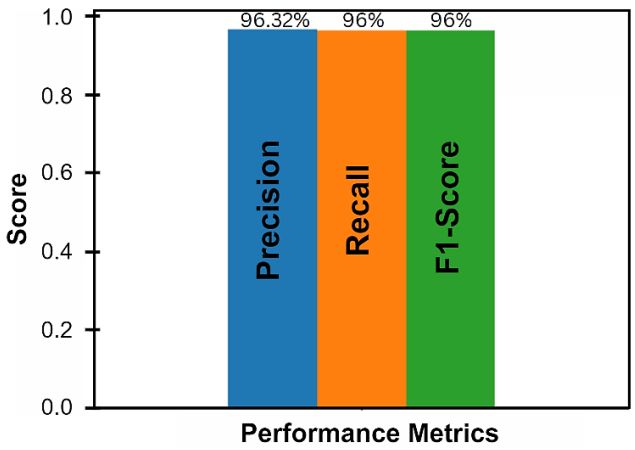

FIGURE 9. Performance metrics for the EDA-Model.

initially untrained state. After the initial drop, the training loss continues to decline at a slower rate, converging to about 0.2 by epoch 50. The validation loss follows a similar pattern, converging to around 0.25 by the end of 50 epochs. The convergence of both training and validation loss curves at a low level suggests that the model is stabilizing and reaching an optimal state of learning without further significant improvements. The convergence of losses at a low level indicates that the model has achieved a stable understanding of the data, and further training may provide diminishing returns in terms of loss reduction. However, if the gap between training and validation losses widens significantly, it could signal overfitting, and regularization techniques or early stopping may be necessary to improve generalization.

For EDA-model, precision value of 96.32% indicates a high accuracy in the positive predictions made by the model (refer to Figure 9). Specifically, it suggests that when the model predicts an outcome as positive, it is correct 96.32% of the time. This high precision is indicative of the model's effectiveness in identifying true positive cases without generating an excessive number of false positives. The recall value of 96% reflects the model's capability to identify all actual positive instances within the dataset. This high recall value demonstrates that the model successfully identifies 96% of all actual positive cases, showcasing its efficiency in capturing nearly all relevant instances. The F1-score, with a value of 96%, is a harmonic mean of precision and recall, providing a single measure that balances the model's accuracy and robustness. High F1-score indicates a balanced model with strong performance in both precision and recall, which is essential in scenarios where both the reliability of the positive predictions and the ability to detect all positives are equally important. The high values across all three metrics suggest that the model is effective and optimized to both detect positive cases and ensure those detections are accurate.

Figure 10 illustrates the confusion matrix for the EDA-model. For class 0, the model exhibits exceptional performance, with 136 instances correctly predicted as belonging

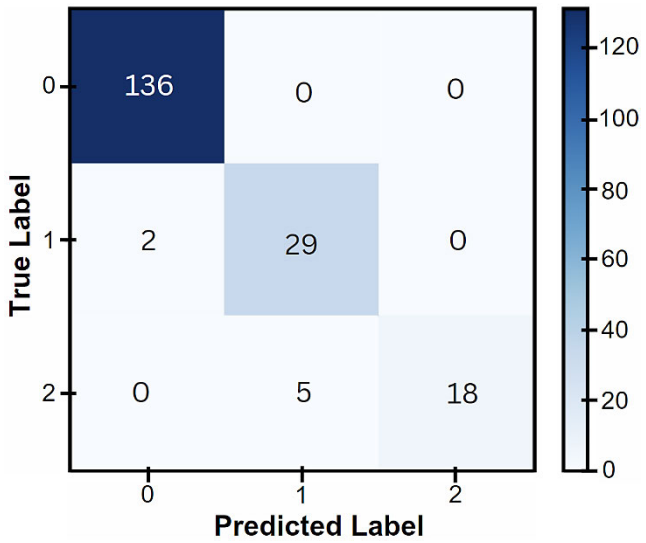

FIGURE 10. Training and validation loss over epochs.

to this class, and no instances misclassified as any other class. This near-perfect classification suggests that the model has effectively learned the distinguishing features of class 0, enabling accurate identification of instances belonging to this class. Regarding class 1, the model demonstrates high precision, with 29 instances correctly predicted as class 1. However, there is some confusion with class 2, as 5 instances of class 2 were incorrectly predicted as class 1.

Additionally, 2 instances of class 1 were misclassified as class 0. This confusion between classes 1 and 2 may indicate similar or overlapping features between these two classes, making it challenging for the model to differentiate them effectively. For class 2, the model correctly predicted 18 instances as belonging to this class. Notably, there were no instances where class 2 was mistaken for class 0, indicating a clear distinction between these two classes. However, 5 instances of class 2 were incorrectly predicted as class 1.

C. EYE-BASED MODEL FOR DROWSINESS DETETCION

A comprehensive dataset comprising a total of 3,951,000 frames for training and 973,800 frames for testing, collected from 30 participants, was employed for training and evaluation purposes. The preprocessing stage involved optimizing pupil recognition identification, PERCLOS, BP, and BR calculations, along with drowsiness labeling. After 100 epochs, the model achieved an approximate accuracy, precision, and recall of 90% on the test set, with a documented test loss of approximately 50%. However, the model struggled to accurately predict instances labeled as 1 or 2 according to the confusion matrix. This difficulty can be attributed to the highly unbalanced nature of the dataset, with 1416 windows of 1800 frames per eye labeled as 0 (active), 396 windows labeled as 1 (possible drowsiness), and 116 windows labeled as 2 (definitely drowsy). The skewed class distribution, predominantly comprising instances of the active (non-drowsy) class, likely hindered the model's ability to effectively predict the minority classes representing varying levels of drowsiness. Please note that the performance plots illustrating the eye tracking-model's behavior are not included in this analysis.

IV. CONCLUSION AND FUTURE IMPLICATIONS

In conclusion, the evaluation of HRV, EDA, and Eye-Models aimed at detecting and classifying driver drowsiness based on physiological indicators has provided valuable insights into their performance and limitations. HRV-Model demonstrated highest accuracy in distinguishing between alert and drowsy drivers, primarily relying on heart rate variability features. Its precision, recall, and F1-score metrics further emphasized its potential in practical utility in real-world scenarios. Similarly, the EDA-Based Model showed significant performance, although it faced difficulties in accurately identifying mild drowsiness instances. This indicates the necessity for refining the model's ability to detect subtle drowsiness states solely based on electrodermal activity features. Conversely, despite achieving a commendable accuracy rate, the Eye-Based Model struggled with accurately pinpointing drowsiness instances, especially positive cases. This challenge was compounded by imbalanced datasets, highlighting the importance of addressing data biases for reliable model performance. As possible solution, future research may explore techniques such as oversampling, under sampling, or synthetic data generation to address this imbalance and potentially improve model performance, especially for minority classes representing varying levels of drowsiness. The models developed in this study have significant potential for real-world applications, particularly in commercial truck driving and fleet management systems. By integrating these drowsiness detection algorithms into existing operator monitoring systems, long-distance drivers can be more effectively monitored for signs of fatigue, thereby enhancing road safety. Despite the promising performance of these models, several limitations were encountered during the study. Technical issues with eye-tracking equipment compromised data quality, while time constraints and session durations might not have captured the full spectrum of fatigue experiences. Additionally, unmeasured external factors like fatigue, stress, and workload could have influenced physiological responses, introducing variability in drowsiness classification. Addressing these limitations and considering external factors are pivotal for refining and deploying drowsiness detection systems effectively in real-world scenarios to improve road safety. Utilizing ensemble machine learning models tailored to different physiological indicators offers a robust approach to comprehensively tackle the complexity of driver drowsiness. Ensemble learning approaches have potential in developing effective driver drowsiness detection systems by synergistically combining multiple machine learning models tailored to diverse neuro-physiological modalities. It can utilize the collective strengths of heterogeneous models, each specialized in capturing distinct drowsiness cues from

various data modalities, such as facial expressions, eye movements, head poses, biosignals, and driving behavior etc. Moreover, ensemble techniques can adapt to varying driving conditions and individual driver characteristics, providing a personalized and context-aware solution for enhancing road safety. By acknowledging and mitigating these limitations, the development of a comprehensive drowsiness detection system holds promise in significantly reducing drowsy driving-related accidents and enhancing driver safety on the roads.

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JOSE ALGUINDIGUE received the M.A.Sc. degree in systems design engineering from the University of Waterloo, in 2023. His research interests include human factors in transportation systems, cognition, and artificial intelligence.

AMANDEEP SINGH received the Ph.D. degree in human factors engineering. He is currently a Post-Doctoral Researcher with the Department of Systems Design Engineering, University of Waterloo. His research interests include human factors in transportation systems, biomechanics, cognitive ergonomics, aging, and human-centered artificial intelligence.

APURVA NARAYAN received the Ph.D. degree in systems design engineering from the University of Waterloo, in 2016. He is currently an Assistant Professor in computer science with Western University. His research interests include data analytics, machine learning, dependable AI systems, decision making under uncertainty, and quantum machine learning.

SIBY SAMUEL received the Ph.D. degree in industrial engineering and operations research (surface transportation human factors) from the University of Massachusetts Amherst, in 2014. He is currently an Assistant Professor with the Department of Systems Design Engineering, University of Waterloo.